A Stem-Routed, Indicator-Driven Viterbi Pipeline for DJ-Set Alignment

Tracklist Engine (`sota_v2`)

April 22, 2026

Abstract

We present *sota_v2*, the alignment algorithm used by Tracklist Engine to assign every track on a 1001tracklists-scraped tracklist to a mix-side time span on a DJ-set recording. The pipeline (i) routes per-ref MERT comparisons through Demucs stems according to the scraped `version_tag`, (ii) infers a monotonic ref-position map with a subsequence-Viterbi DP (replacing argmax, which is non-monotonic and produces impossible descending brackets), (iii) composes an emission score out of three time-series indicators borrowed from technical analysis — MACD, ATR-normalised, a cross-sectional z-score within a variant-group, and a pre-cue persistence baseline — (iv) runs a per-universe $(K+1)$-state Viterbi with a SILENCE state, Phase-5 fingerprint anchors, and a Phase-6 scale-gated cross-universe exclusion mask, and (v) snaps decoded ref endpoints to canonical cue-detr cues computed once per song. On the BB11 ground-truth fixture the pipeline reaches mean mix IoU 0.891 over the five hand-annotated refs. We report scale failure modes at 119 refs and the mitigations we added.

1 Problem statement

Given (a) a DJ-set recording M sampled at 44.1 kHz with a pre-computed beat-grid $\{t_0, t_1, \ldots, t_{T-1}\}$ of measure centres, and (b) a tracklist $\mathcal{R} = \{r_1, \ldots, r_N\}$ scraped from 1001tracklists, where each $r \in \mathcal{R}$ is associated with a downloaded reference audio file, its own measure grid, a scraped cue time c_r, and a `version_tag` indicating which variant the DJ is playing (*acappella*, *instrumental*, or *full*), we seek a mapping

$$\Phi : \mathcal{R} \to \big(\mathbb{R} \times \mathbb{R} \times \mathbb{R} \times \mathbb{R}\big) \cup \{\perp\}$$

assigning each ref either a tuple $(s_r^{\text{mix}}, e_r^{\text{mix}}, s_r^{\text{ref}}, e_r^{\text{ref}})$ describing its mix-side play span and the ref-side region that was played, or $\perp$ (did not play). Alignments are evaluated against hand-annotated ground-truth fixtures using span-IoU.

2 Notation

- T — number of mix measures; N_r — number of measures of ref r.
- $\mathbf{E}^{\text{mix}} \in \mathbb{R}^{T\times d}$, $\mathbf{E}^{r} \in \mathbb{R}^{N_r\times d}$ — per-measure MERT embeddings on the mix side and the r side respectively.
- $S_r \in \mathbb{R}^{N_r\times T}$ — cosine similarity between every ref-measure and every mix-measure of a pairing.
- $\sigma_r(t) \coloneqq \max_j S_r[j,t]$ — per-mix-measure max-sim score for ref r.
- $U(r) \in \{\text{acap}, \text{instr}, \text{full}\}$ — the universe (mutual-exclusion group) of r.
- $\mathcal{U}_u = \{r : U(r) = u\}$ — the refs in universe u.

3 Pipeline overview

Algorithm 1 gives the end-to-end flow. Each numbered stage is expanded in the following sections.

Algorithm 1 SotaAlignSet(set_id)

$\mathcal{R}, M \leftarrow$ load refs with audio+measures and the set audio
for $r \in \mathcal{R}$ **do**
  stem-route: pick ref-side and mix-side audio paths per $U(r)$
  $\mathbf{E}^r, \mathbf{E}^{\text{mix}} \leftarrow$ MERT-embed per measure (Sec. 4)
  $S_r \leftarrow \mathbf{E}^r \cdot (\mathbf{E}^{\text{mix}})^\top$ ▷ cosine similarity
  $\sigma_r \leftarrow \max_{\text{axis}=0}(S_r)$
  $\pi_r \leftarrow$ RefPositionViterbi(S_r) ▷ Sec. 5
end for
$A \leftarrow$ LoadFingerprintAnchors(set_id) ▷ Phase 5
$X \leftarrow$ BuildFullExclusionMask(A, $\mathcal{R}$) with coverage cap ▷ Phase 6
for $u \in \{\text{acap}, \text{instr}, \text{full}\}$ **do**
  $P_u \leftarrow$ ViterbiUniverse($\mathcal{U}_u, \sigma, A, X_{u \neq \text{full}}$) ▷ Sec. 9
  $P_u \leftarrow$ CleanPath(P_u) ▷ Sec. 10
end for
for $r \in \mathcal{R}$ **do**
  extract first run (s, e) of r in $P_{U(r)}$; read endpoints via π_r
  snap $(s_r^{\text{ref}}, e_r^{\text{ref}})$ to nearest canonical cue; apply mix-side shift
end for
persist to `set_section_alignment` with `confidence_source='sota_v2'`

4 Signal extraction

4.1 Stem routing

Let stem(u) denote the audio stream a ref in universe u should be compared against:

$$\text{stem}(u) = \begin{cases} \text{vocals} & u = \text{acap} \\ \text{instrumental} & u = \text{instr} \\ \text{full audio} & u = \text{full} \end{cases}$$

We apply the *same* stem(u) on both sides of the pairing: the mix is demucs-separated once per set and the relevant stem embedded, and each ref is embedded from its own corresponding stem. Routing an acappella-tagged ref against the mix's vocals stem suppresses the instrumental bed that would otherwise dominate MERT's cosine similarity.

4.2 MERT embeddings and cosine similarity

Each stem-routed audio file is segmented according to its measure grid; the MERT v1 encoder produces a d-dimensional embedding per measure after mean-pooling across the token axis. We use the layer-5 average as a fixed feature space (determined to be the most alignment-informative on a held-out subset). Similarity between ref-measure j and mix-measure t is the cosine

$$S_r[j, t] = \frac{\mathbf{E}_j^r \cdot \mathbf{E}_t^{\text{mix}}}{\|\mathbf{E}_j^r\| \, \|\mathbf{E}_t^{\text{mix}}\|}.$$

5 Ref-position Viterbi

Given S_r, we want a monotonic map $\pi_r : \{0, \ldots, T-1\} \to \{0, \ldots, N_r-1\}$ answering *where in ref r is the DJ at mix time t?*. Argmax over rows violates monotonicity — identical choruses at different ref times confuse it — and produces the pathological brackets $[s^{\text{ref}} > e^{\text{ref}}]$ we observed pre-Viterbi. We instead solve a subsequence-DTW-style shortest-path problem on a $T \times N_r$ cost lattice:

$$\mathcal{C}_t[j] = (1 - S_r[j,t]) + \min_{j'} \Big(\mathcal{C}_{t-1}[j'] + \tau(j - j')\Big), \tag{1}$$

$$\tau(\delta) = \begin{cases} 0 & \delta \in \{0,1\} \quad \text{(stay or advance)} \\ 0.3\,\delta & \delta \geq 2 \quad \text{(speed-up skip)} \\ 2.0\,|\delta| & \delta < 0 \quad \text{(loop-back)} \end{cases} \tag{2}$$

The large backward cost forbids path descent in almost all cases, eliminating the invalid descending-bracket failure mode. The optimisation is restricted to a band $|j - j'| \leq 8$ for compute. The backpointer lattice $\mathcal{B}$ is recovered in $O(T)$ to yield π_r.

Algorithm 2 RefPositionViterbi(S_r)

$\mathcal{C}_0[j] \leftarrow 1 - S_r[j, 0]$ for all j
for $t = 1, \ldots, T-1$ **do**
  for $j = 0, \ldots, N_r - 1$ **do**
    $(\mathcal{C}_t[j], \mathcal{B}_t[j]) \leftarrow \min_{|j'-j|\leq 8} \big(\mathcal{C}_{t-1}[j'] + \tau(j-j')\big) + (1 - S_r[j,t])$
  end for
end for
Backtrace from $\arg\min_j \mathcal{C}_{T-1}[j]$ through $\mathcal{B}$ to recover π_r

6 Emission signals: time-series indicators as alignment features

Rather than consume σ_r directly as a score, we compose it with three orthogonal signals borrowed from technical analysis of financial time-series. Each captures a distinct notion of evidence that ref r is currently playing.

6.1 MACD histogram of similarity momentum

Let $\text{EMA}_n(x)$ denote the exponential moving average with span n. The MACD of σ_r is

$$\text{MACD}_r(t) = \text{EMA}_{12}(\sigma_r)(t) - \text{EMA}_{26}(\sigma_r)(t),$$

the signal line is $\text{Signal}_r(t) = \text{EMA}_9(\text{MACD}_r)(t)$, and the *histogram*

$$h_r(t) = \text{MACD}_r(t) - \text{Signal}_r(t)$$

fires sharply positive at *entries* (similarity rises faster than its recent trend) and negative at *exits* (the ref stops). It is the most informative signal for boundary detection.

To put different refs on a comparable scale we divide by the Wilder-smoothed average true range of σ_r, $\text{ATR}_{14}(\sigma_r)$, and clip:

$$\tilde{h}_r(t) = \text{clip}\left(\frac{h_r(t)}{\max(\text{ATR}_{14}(\sigma_r)(t), 10^{-4})}, -3, 3\right).$$

6.2 Cross-sectional z-score within universe

At each mix measure t we ask: of all refs in the same mutual-exclusion group $\mathcal{U}_u$, which is the *relative* winner?

$$\mu_u(t) = \frac{1}{|\mathcal{U}_u|} \sum_{r \in \mathcal{U}_u} \sigma_r(t), \quad s_u(t) = \sqrt{\tfrac{1}{|\mathcal{U}_u|} \sum_{r \in \mathcal{U}_u} (\sigma_r(t) - \mu_u(t))^2},$$

$$z_r(t) = \frac{\sigma_r(t) - \mu_u(t)}{\max(s_u(t), 10^{-6})}, \quad r \in \mathcal{U}_u.$$

This is the "max-of-acappellas at each time" idea formalised: picking the acappella with the strongest relative evidence. For singleton universes ($|\mathcal{U}_u| = 1$) we fall back to a rolling temporal z-score of σ_r against its own 40-measure history, because a ref with no in-universe competitors would otherwise have a constant-zero z and lose every winner-takes-all contest against SILENCE.

6.3 Pre-cue persistence baseline

Because the DJ's scraped cue c_r is an approximate left-bound, we estimate ref r's *silence baseline* as the mean similarity before the cue:

$$b_r = \text{mean}_{\{t:t<c_r\}} \sigma_r(t),$$

falling back to the 10$^{\text{th}}$ percentile of σ_r if fewer than three pre-cue measures exist. The persistence score

$$p_r(t) = \text{clip}\big(8 \cdot (\sigma_r(t) - b_r),\ -2,\ 2\big)$$

stays positive for the *full* duration of a play (unlike MACD's histogram, which fires only at entries/exits), holding the Viterbi in-state during long sustained plays.

6.4 Composite emission score

The per-ref emission signal consumed by the per-universe Viterbi is a weighted sum:

$$\mathcal{E}_r(t) = 0.5\, z_r(t) + 1.0\, \tilde{h}_r(t) + 1.5\, p_r(t). \tag{3}$$

The weights were tuned on BB11 ground-truth and not re-tuned since.

7 Fingerprint anchors (Phase 5)

We pre-compute chromaprint fingerprints for every track and scan the mix against them. A fingerprint hit is a tuple $(t^{\text{mix}}_{\text{start}}, t^{\text{mix}}_{\text{end}}, \text{score}, \text{matched\_track\_id})$. Single hits are noisy (a lone 0.7-score hit is nearly meaningless), but *clusters* of hits within a small window are strong positive evidence. For each ref r with track_id τ we build a boolean anchor mask $\alpha_r \in \{0,1\}^T$:

$$\alpha_r(t) = \mathbb{1}\big[\,\#\{\text{hits of } \tau \text{ with centre within } \pm 10\,\text{s of } t_t,\ \text{score} \geq 0.65\} \geq 2\,\big].$$

Where $\alpha_r(t) = 1$, the per-universe Viterbi receives a -1.5 emission-cost bonus for state r, reinforcing the decode at density-confirmed measures.

8 Full-track exclusion with scale-aware cap (Phase 6)

When a *full*-variant ref is fingerprint-confirmed, layering is unlikely: the DJ has swapped to playing a whole track, not overlaying an acappella on top. We construct a cross-universe exclusion mask

$$X(t) = \bigvee_{r:U(r)=\text{full}} \alpha_r(t).$$

At measures where $X(t) = 1$, the emission cost for every non-silence state in the *acappella* and *instrumental* universes is set to 10^6, forcing those universes to SILENCE.

Scale failure mode. At small $|\mathcal{R}|$ the union X covers a small fraction of the mix ($\approx 15\%$ on BB11's 5-ref GT). At $|\mathcal{R}| = 119$ we observed 72% coverage, hard-masking genuine acappella plays and regressing Gnash's IoU from 0.857 to 0.27.

Mitigation. We apply X only if

$$\frac{1}{T}\sum_t X(t) \le 0.40.$$

Above the cap, X is discarded and the within-universe Viterbi is allowed to decide. The cap preserves the validated 5-ref behaviour while avoiding the hard-mask regression at scale.

9 Per-universe K+1-state Viterbi

Within each universe u we decode a Viterbi with state space $\{0, 1, \ldots, |\mathcal{U}_u|-1\} \cup \{\texttt{SIL}\}$ (one state per ref in the universe plus a single silence state). Emission cost for ref state i at mix measure t is

$$\mathcal{E}_i^{\text{cost}}(t) = -\mathcal{E}_{r_i}(t) - 1.5 \cdot \alpha_{r_i}(t) + 10^6 \cdot \mathbb{1}[X(t) \wedge u \neq \text{full}],$$

and SILENCE is constant at -1.2. We apply a hard left-boundary: $\mathcal{E}_i^{\text{cost}}(t) = +\infty$ for t such that $t_t < c_{r_i}$. Transition costs are

	from			
to	i	$j \neq i$	`SIL`	(note)
i (self-loop)	0	3.0	0.4	enter-from-silence is 0.4
$j \neq i$	3.0	—	0.4	cross-ref forced via SILENCE
`SIL`	0.1	0.1	0	exit-to-silence is 0.1

The 3.0 cross-ref cost makes direct ref-to-ref transitions strictly worse than going through SILENCE, producing at most one active ref per universe per measure. The full Viterbi forward pass is $O(T \cdot (|\mathcal{U}_u| + 1)^2)$.

10 Path cleanup

Raw Viterbi paths are fragmented by short incidental SILENCE wins within real plays (1–3 measures). Cleanup per ref i performs:

1. **Merge.** Extract all contiguous runs of state i. Merge two consecutive runs iff the gap is ≤ 10 measures *and* every cell in the gap is SILENCE.
2. **Cue-anchored selection.** Pick the *earliest* merged run of duration ≥ 5 measures whose start is within ± 80 s of the scraped cue c_{r_i}. All other runs (later re-entries, spurious short fires, far-from-cue) are wiped to SILENCE.

3. **Drop.** If no run qualifies, ref i produces $\Phi(r_i) = \perp$.

The earliest-near-cue rule encodes the DJ-play prior: each ref is played once, starting near its scraped cue. Later re-entries are almost always ref-collision artefacts (a competing ref's signal leaking into this ref's state).

11 Endpoint extraction and canonical-cue snap

From the cleaned run (s, e) of ref r we read the ref-position Viterbi path π_r at both boundaries. To tolerate single-measure noise we take the median of the first and last three measures of π_r within the run:

$$j_r^{\text{lo}} = \text{median}\big(\pi_r[s : s+3]\big), \quad j_r^{\text{hi}} = \text{median}\big(\pi_r[e-3 : e]\big).$$

These map to ref-time seconds $t_r^{\text{ref,lo}}$, $t_r^{\text{ref,hi}}$ via the ref-side measure centres.

Canonical cue snap. Each song has a canonical cue list $\mathcal{K}_\tau$ computed *once* on its original `variant_tag='original'` full-song audio at cue-detr sensitivity 0.5, stored in `canonical_track_cue_points` keyed by `track_id`. All variants (acappella, instrumental, full, remix) share the same cue list because the underlying song structure is the same. We snap

$$\tilde{t}_r^{\text{ref,lo}} = \arg\min_{k \in \mathcal{K}_\tau} |k - t_r^{\text{ref,lo}}|, \quad \tilde{t}_r^{\text{ref,hi}} = \arg\min_{k \in \mathcal{K}_\tau} |k - t_r^{\text{ref,hi}}|.$$

Assuming a tempo factor of ≈ 1, the mix-side span is shifted by the same delta:

$$\tilde{s}_r^{\text{mix}} = s_r^{\text{mix}} - (t_r^{\text{ref,lo}} - \tilde{t}_r^{\text{ref,lo}}), \quad \tilde{e}_r^{\text{mix}} = e_r^{\text{mix}} - (t_r^{\text{ref,hi}} - \tilde{t}_r^{\text{ref,hi}}).$$

We *skip* the snap for u = full refs, because cue-detr on a full-band track fires a lot and the nearest cue frequently lies well inside the real play span, regressing the outer-boundary IoU (verified: snapping Antoine on BB11 regressed its IoU from 0.950 to 0.826).

12 Evaluation

12.1 Ground-truth fixture

We annotate a DJ-set against a user-maintained YAML fixture at `tests/fixtures/<set_id>_ground_truth.ya` with hand-verified $(s^{\text{mix}}, e^{\text{mix}})$ for every played track, plus optional `ref_segments` for loops and cut-ups. The primary mix-side metric is span-IoU

$$\text{IoU}_r = \frac{|[\tilde{s}_r^{\text{mix}}, \tilde{e}_r^{\text{mix}}] \cap [s_r^{\text{mix,gt}}, e_r^{\text{mix,gt}}]|}{|[\tilde{s}_r^{\text{mix}}, \tilde{e}_r^{\text{mix}}] \cup [s_r^{\text{mix,gt}}, e_r^{\text{mix,gt}}]|}.$$

12.2 BB11 five-reference results

On Big Bootie Volume 11 with the five hand-annotated refs the pipeline achieves:

Ref (variant)	Mix IoU (raw)	Mix IoU (with snap)	snap Δ
Bastille — Good Grief (instr)	0.966	0.977	+0.011
Fray — How to Save a Life (acap)	0.960	1.000	+0.040
Carly Rae Jepsen — Call Me Maybe (acap)	0.625	0.625	0
Gnash — I Hate U, I Love U (acap)	0.857	0.905	+0.048
Antoine Delvig — Blondies (full, *no* snap)	0.950	0.950	0
mean (5 matched refs)	**0.872**	**0.891**	**+0.019**

The mean is 0.891 when the cue-snap is promoted to the final prediction. Argmax-based ref-position inference (the pre-Viterbi baseline) produces impossible descending brackets for two of the five refs and means 0.751.

12.3 Ablations

Configuration	Mean mix IoU
Full pipeline	**0.891**
− cue snap	0.872
− ref-position Viterbi (argmax fallback)	0.751
− stem routing (all refs vs. full mix)	catastrophic on acappella refs
− MACD term in emission	severe drop on entry/exit boundaries
− cross-sectional z	winner-selection breaks within universe
− persistence term	short runs lose to SILENCE mid-play
+ BPM penalty in emission	regresses Gnash 0.857 → 0.381 (DJs tempo-shift)

13 Discussion: scale failure modes

Applying *sota_v2* to the full BB11 tracklist (119 refs, vs. the 5-ref GT subset) dropped mean IoU to 0.629 before mitigation. We identified three failure modes, in descending order of impact:

1. X **over-masking (dominant).** The full-track exclusion union grows with $|\mathcal{U}_{\text{full}}|$; at 119 refs it covered 72% of the mix, hard-forcing acappella and instrumental universes to SILENCE across genuine play windows. *Mitigation*: the coverage cap in Sec. 8 — if $\frac{1}{T}\sum_t X(t) > 0.40$, disable X entirely for the set.
2. **Version-tag mis-classification.** `_normalise_version_tag` is a keyword regex; refs whose scraped metadata lacks “acap”, “instr”, or “dub” default to *full*, over-populating $\mathcal{U}_{\text{full}}$ and feeding (1). For the GT-validated BB11 subset we maintain a hand-curated override map.
3. **Cross-sectional z dilution.** μ_u, s_u are computed over the full universe. With $|\mathcal{U}_u| \approx 40$ per universe the signal-to-noise of each individual z_r degrades. An open direction is a rolling temporal z per ref instead of (or combined with) the cross-sectional one.

14 Implementation and reproduction

The canonical entry point is `audio_pipeline/alignment/sota.py`; the Viterbi primitives and indicator math live in `audio_pipeline/alignment/indicators_debug.py`. To regenerate the BB11 alignment and evaluate:

```
venvs/audio/bin/python -m audio_pipeline.alignment.sota --set-id 2nvzlh2k
venvs/audio/bin/python -m audio_pipeline.alignment.eval \
    --db data/db/music_database.db
```

For fast iteration against only the GT-fixture refs, pass `--track-ids g8gtgdx,26b4gz6f,4gy6y1p,2m5wh0t5,r`

15 Open problems

1. **Version-tag audio-level verification.** Replace the scraped-metadata regex with a classifier that listens to the ref audio (vocals-to-total energy ratio, duration) and confirms the variant.

2. **Scale-aware z.** Rolling temporal z or robust location/scale (median / MAD) per ref rather than pooled cross-sectional moments, to stabilise emission signals at large $|\mathcal{U}_u|$.
3. **Soft full-exclusion.** Replace the hard 10^6 mask with a continuous penalty weighted by the density / confidence of the triggering fingerprint anchor — recovers (1)'s benefit at small scale without the pathological behaviour at large scale.
4. **Loops and cut-ups.** The current pipeline emits a single $(s^{\text{ref}}, e^{\text{ref}})$ bracket per ref. GT fixtures admit `ref_segments` for loops; decoding the ref-position Viterbi into multiple runs and reporting each is the natural extension.